

ELEC 221 Lecture 08

DT Fourier series and filters

Tuesday 6 October 2026

Announcements

- Quiz 4 today
- Tutorial Assignment 2 due Friday 23:59

Last time

Fourier synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

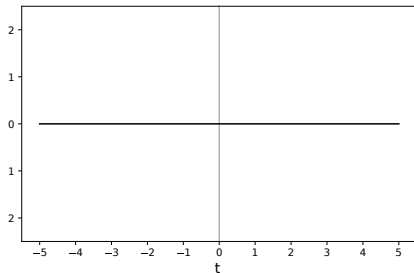
Fourier analysis equation:

$$c_k = \frac{1}{T} \int_T e^{-jk\omega t} x(t) dt$$

Last time

We evaluated the Fourier series coefficients of a square wave:

$$x(t) = \begin{cases} 1, & 0 \leq t < \pi, \\ -1, & \pi \leq t < 2\pi \end{cases}$$



$$x(t) = \sum_{k=1}^{\infty} b_k \sin(kt), \quad b_k = \begin{cases} 0, & k \text{ is even} \\ 4/k\pi, & k \text{ is odd} \end{cases}$$

Last time

We determined how Fourier series coefficients transform.

Superposition of two signals with same ω :

$$\begin{aligned}x(t) &\leftrightarrow a_k, & y(t) &\leftrightarrow b_k \\z(t) = Ax(t) + By(t) &\Rightarrow c_k = Aa_k + Bb_k\end{aligned}$$

Time shift

$$x(t) \leftrightarrow c_k \quad \Rightarrow \quad x(t - t_0) \leftrightarrow e^{-jk\omega t_0} c_k$$

Time scale

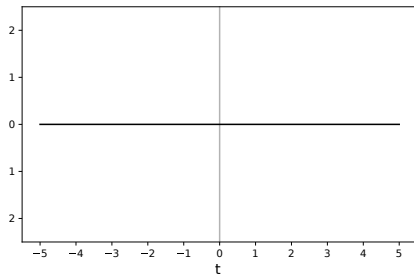
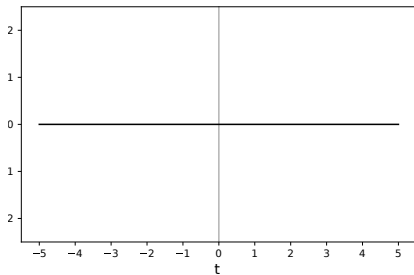
$$x(t) \leftrightarrow c_k \quad \Rightarrow \quad x(\alpha t) \leftrightarrow c_k, \quad x(\alpha t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\alpha\omega t}$$

Multiplication leads to convolution:

$$z(t) = x(t)y(t) \quad \leftrightarrow \quad c_k = \sum_{\ell=-\infty}^{\infty} a_{\ell} b_{k-\ell}$$

Last time

We used the frequency response to define some simple filters.



Today

Learning outcomes:

- Determine the frequency response of an RC filter circuit
- Compute the fundamental period and frequency of DT signals
- Evaluate Fourier series coefficients of DT signals
- Define and distinguish between finite-impulse-response (FIR) and infinite-impulse-response (IIR) filters in DT

Example: Lowpass filters in practice

Some filters made of physical components are described by differential equations.

Example: a lowpass filter using a resistor and a capacitor.

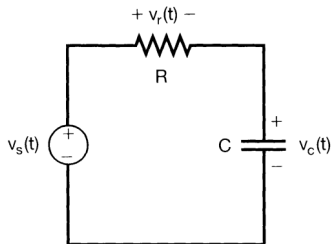
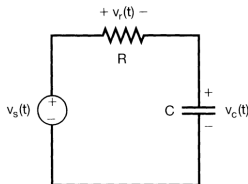


Image credit: Oppenheim chapter 3.10.

Example: lowpass filters in practice

What is the voltage across the capacitor if $v_s = e^{j\omega t}$?



Can derive two expressions for the current, using the resistor and capacitor respectively:

$$i(t) = \frac{v_s(t) - v_c(t)}{R}$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

Example: lowpass filters in practice

Put these together to form a differential equation:

$$C \frac{dv_c(t)}{dt} = \frac{v_s(t) - v_c(t)}{R}$$

$$RC \frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

$$RC \frac{d}{dt}(H(j\omega)e^{j\omega t}) + H(j\omega)e^{j\omega t} = e^{j\omega t}$$

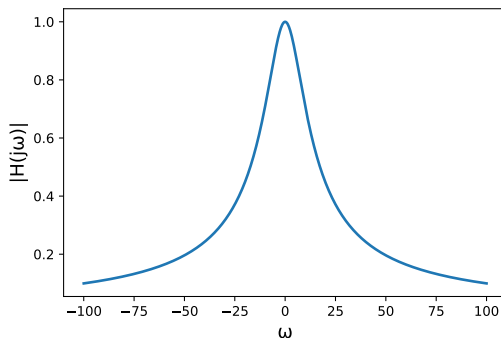
$$RCj\omega H(j\omega)e^{j\omega t} + H(j\omega)e^{j\omega t} = e^{j\omega t}$$

$$(RCj\omega + 1)H(j\omega)e^{j\omega t} = e^{j\omega t}$$

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

Example: lowpass filters in practice

Results in the following frequency response (setting $RC = 0.1$):



Adjusting the value of RC controls the frequency response. Increasing RC cuts off more frequencies.

DT complex exponential signals

Recall our CT representation of complex exponential signals:

$$x(t) = Ce^{\alpha t}$$

where α could be real or complex.

In DT, we write

$$x[n] = C\beta^n \quad (= Ce^{\alpha n})$$

where β can be real or complex.

DT complex exponential signals

Case: $x[n] = C\beta^n$, where β is real.

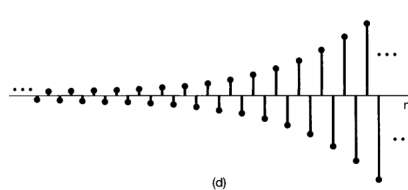
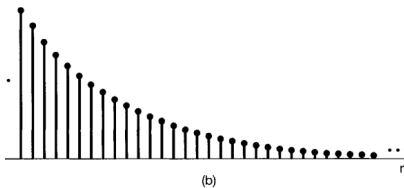
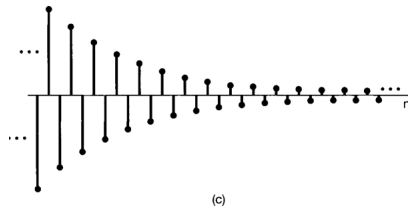
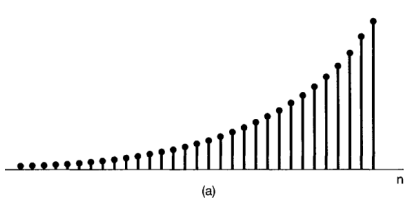
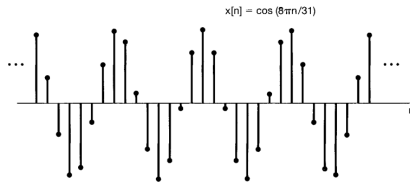
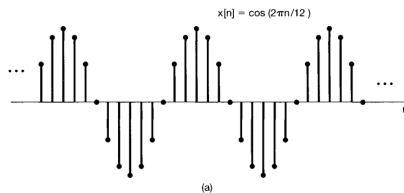


Image credit: Oppenheim chapter 1.3.

DT complex exponential signals

Case: $x[n] = C\beta^n$, where β is purely complex:

$$x[n] = Ce^{j\omega n} = C \cos(\omega n) + Cj \sin(\omega n)$$



Frequency and period of DT complex exponential signals

While these might look similar to their CT counterparts, there is a **very important difference** relating to frequency.

In CT,

$$x(t) = A \cos(\omega t + \theta), \quad x(t) = Ce^{j\omega t}$$

This is periodic with period $T = 2\pi/\omega$

The bigger the frequency (ω) gets, the faster it oscillates!

Frequency and period of DT complex exponential signals

Exercise: consider the DT signal

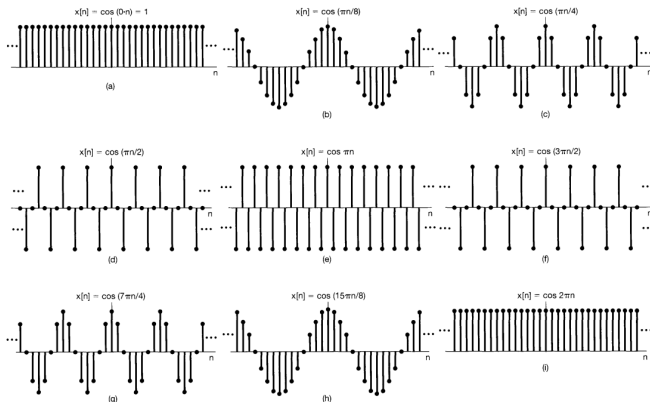
$$x[n] = e^{j\omega n}$$

Does bigger ω always mean faster oscillation? If yes, why? If no, when does it stop getting faster?

$$\begin{aligned} x[n] &= e^{j(\omega+2\pi)n} \\ &= e^{j\omega n} e^{j2\pi n} \\ &= e^{j\omega n} \end{aligned}$$

Frequency and period of DT complex exponential signals

For a DT signal with frequency ω , the signals with frequencies $\omega \pm 2n\pi$ are the same!



Only need to consider ω in the range 0 to 2π (or $-\pi$ to π).

Frequency and period of DT complex exponential signals

Exercise: What are the fundamental periods of

$$x(t) = \cos(3t), \quad \text{and} \quad x[n] = \cos(3n)$$

Frequency and period of DT complex exponential signals

Suppose the period is N :

$$\begin{aligned}x[n] &= e^{j\omega(n+N)} \\ &= e^{j\omega n} e^{j\omega N}\end{aligned}$$

This implies

$$e^{j\omega N} = 1 \quad \rightarrow \quad \omega N = 2\pi m, \quad m \in \mathbb{Z}$$

$\omega/2\pi$ must be rational for the signal to be periodic.

Frequency and period of DT complex exponential signals

Exercise: what is the fundamental period of

$$x[n] = \cos(5\pi n/6) + \sin(2\pi n/3)$$

For the CT equivalent, $T = 2\pi/\omega = 12/5$, but this is not a valid n . But after 5 periods in CT, we are at $n = 12$ which is a whole number, so the fundamental period is $N = 12$.
For the second term, $T = 3$ which is a valid N . So, the fundamental period of the whole signal is $N = 12$.

Harmonics of DT complex exponential signals

What about harmonics?

$$x_k[n] = e^{jk\omega n} = e^{jk\frac{2\pi n}{N}}, \quad k \in \mathbb{Z}$$

In CT we had an infinite number of these. What about DT?

Once we reach $k = N$, we get back to our original signal... so have only N distinct harmonics

$$x_k[n] = e^{jk\frac{2\pi n}{N}}, \quad k \in 0, \dots, N-1$$

DT signals and LTI systems

Consider a system with impulse response $h[n]$ and DT signal $x_m[n] = e^{jm\omega n}$. Use the convolution sum:

$$\begin{aligned}y[n] &= \sum_{k=-\infty}^{\infty} x[n-k]h[k] \\&= \sum_{k=-\infty}^{\infty} e^{jm\omega(n-k)}h[k] \\&= e^{jm\omega n} \sum_{k=-\infty}^{\infty} e^{-jm\omega k}h[k] \\&= x_m[n] \cdot H(e^{jm\omega})\end{aligned}$$

DT Fourier series

If we know how a system responds to complex exponential signals, we can learn its response to signals expressed in terms of them.

We need a Fourier series representation of DT signals:

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk\omega n}$$

How do we find the c_k ?

DT Fourier coefficients

Leverage the following identity about complex numbers:

$$\sum_{k=0}^{N-1} e^{jk\frac{2\pi n}{N}} = \begin{cases} N & \text{if } k = 0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$$

We will multiply on both sides, and sum.

DT Fourier coefficients

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi n}{N}}$$

$$\sum_{n=0}^{N-1} e^{-jm \frac{2\pi n}{N}} x[n] = \sum_{n=0}^{N-1} e^{-jm \frac{2\pi n}{N}} \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi n}{N}}$$

$$= \sum_{k=0}^{N-1} c_k \sum_{n=0}^{N-1} e^{j(k-m) \frac{2\pi n}{N}}$$

$$= c_m N$$

DT Fourier coefficients

DT Fourier synthesis equation

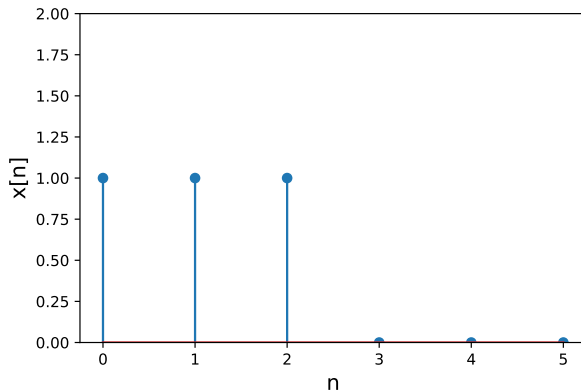
$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi n}{N}}$$

DT Fourier analysis equation

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk \frac{2\pi n}{N}} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi n}{N}}$$

Exercise: the DT square wave

Compute the Fourier coefficients of this signal:



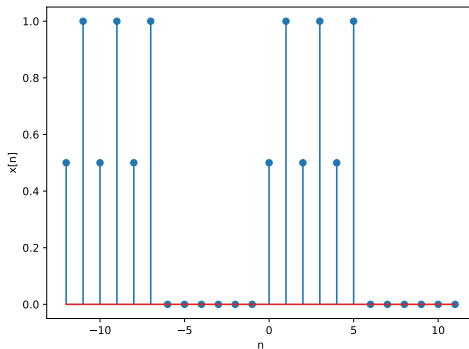
Exercise: the DT square wave

Properties of DT Fourier coefficients

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$\left. \begin{array}{l} x[n] \\ y[n] \end{array} \right\} \begin{array}{l} \text{Periodic with period } N \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/N \end{array}$	$\left. \begin{array}{l} a_k \\ b_k \end{array} \right\} \begin{array}{l} \text{Periodic with} \\ \text{period } N \end{array}$
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a_{-k}^*
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m} a_k \quad \left(\begin{array}{l} \text{viewed as periodic} \\ \text{with period } mN \end{array} \right)$
Periodic Convolution	$\sum_{r=(N)} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=(N)} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^n x[k] \quad \left(\begin{array}{l} \text{finite valued and periodic only} \\ \text{if } a_0 = 0 \end{array} \right)$	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})} \right) a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$\left\{ \begin{array}{l} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{array} \right.$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\left\{ \begin{array}{ll} x_e[n] = \mathcal{E}\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{O}\{x[n]\} & [x[n] \text{ real}] \end{array} \right.$	$\left\{ \begin{array}{l} \Re\{a_k\} \\ j\Im\{a_k\} \end{array} \right.$

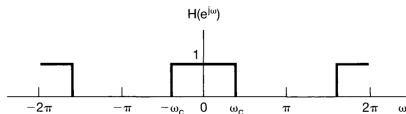
How would we compute the Fourier coefficients for this signal?



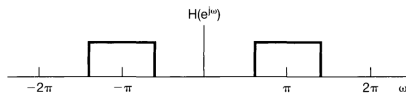
Exercise: DT Fourier coefficients

DT filters

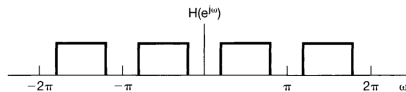
Recall that in DT, the frequency increases up until $\omega = \pi$, then decreases as it approaches 2π .



(a)



(b)



(c)

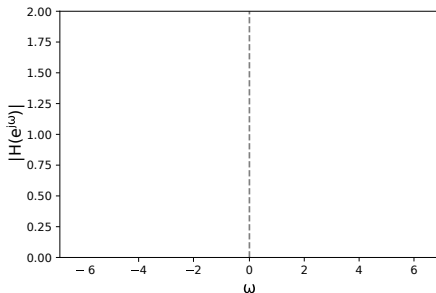
Image credit: Oppenheim chapter 3.9.

DT filters

Example: consider the DT filter with frequency response

$$H(e^{j\omega}) = \begin{cases} \frac{3}{2}, & \frac{\pi}{6} \leq |\omega| \leq \frac{\pi}{4} \\ 1, & \frac{\pi}{2} \leq |\omega| \leq \frac{3\pi}{4} \\ 0, & \text{otherwise} \end{cases}$$

Sketch below:



DT filters

There are two major categories of DT filters:

- ① Infinite impulse response (IIR)
- ② Finite impulse response (FIR)

DT filters (IIR)

Example: we can generate a low- or high-pass filter using a system described by a first-order difference equation.

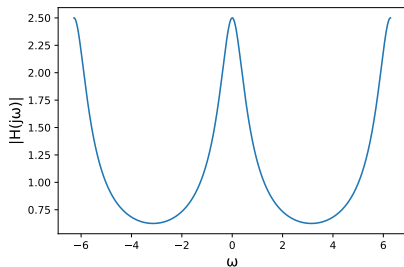
$$\begin{aligned}y[n] - ay[n-1] &= x[n] \\H(e^{j\omega})e^{j\omega n} - aH(e^{j\omega})e^{j\omega(n-1)} &= e^{j\omega n} \\H(e^{j\omega})(1 - ae^{-j\omega})e^{j\omega n} &= e^{j\omega n} \\H(e^{j\omega}) &= \frac{1}{1 - ae^{-j\omega}}\end{aligned}$$

This system has infinite impulse response:

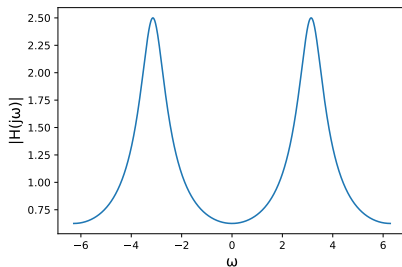
$$h[n] = a^n u[n]$$

DT filters (IIR)

Case: $a > 0$



Case: $a < 0$



DT filters (FIR)

An example of a DT filter with finite impulse response is the moving average with window of size M :

$$y[n] = \frac{1}{2M+1} \sum_{k=-M}^M x[n-k]$$

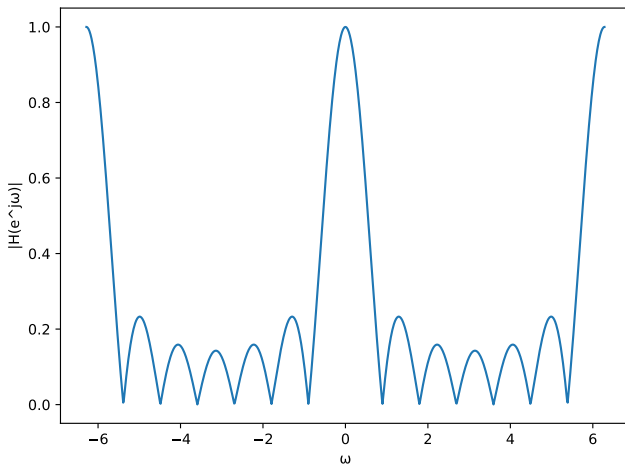
Impulse response: set $x[n] = \delta[n]$

$$\begin{aligned} h[n] &= \frac{1}{2M+1} \sum_{k=-M}^M \delta[n-k] \\ &= \frac{1}{2M+1} \end{aligned}$$

This is clearly finite (as a result: these filters are stable systems).

DT filters (FIR)

Example when $M = 3$



For next time

Content:

- The CT Fourier transform

Action items:

- ① Tutorial Assignment 2 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 3.5-3.12
- Suggested problems: 3.2, 3.10-3.12, 3.14, 3.16, 3.17, 3.19, 3.20 3.23, 3.27-3.32, 3.36, 3.39
- For next class: Oppenheim 4.1-4.3